

ZIGEN: A Windowing System Enabling Multitasking Among 3D and 2D Applications in Immersive Environments

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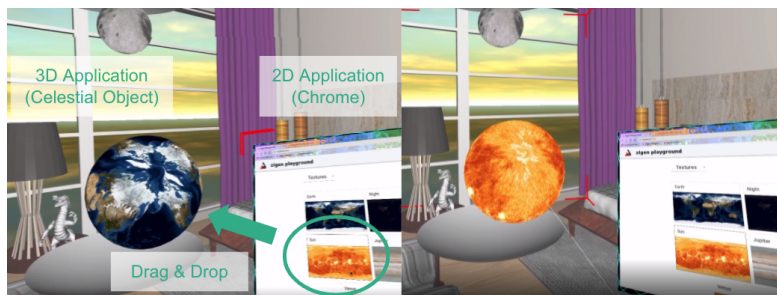


Figure 1: By drag-and-drop, texture is passed from an unmodified 2D application to a 3D application, and the object in the 3D application is changed.

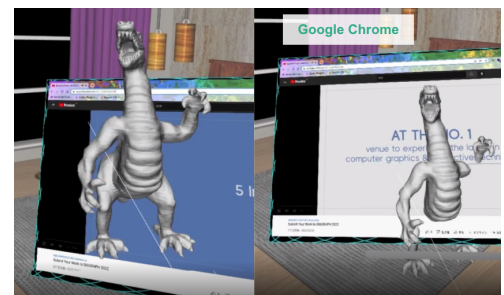


Figure 2: A 2D application (Google Chrome) and a 3D application are composited.

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1 INTRODUCTION

Windowing systems are an integral part of the modern desktop environment. Thanks to them, we are able to use multiple graphical user interface (GUI) applications from different developers simultaneously on a single display. They not only display multiple applications as overlapping windows, but also abstract input events such as keyboard and mouse events, and define the concept of focus which is the destination of these events, so that input events are transferred only to the application intended by the user. Furthermore, they define protocols such as drag-and-drop so that data can be passed between applications even if they are developed by

different developers. These windowing system features provide a sophisticated multitasking space in the 2D desktop environment.

However, in current immersive 3D user interfaces, basically one main application dominates a user's entire field of view. Therefore, it is not possible to install multiple applications of the user's choice and use them simultaneously to create the user's own 3D workspace. OpenXR's overlay feature [The Khronos Group Inc. 2022] allows auxiliary applications to be displayed together with the main application. However, it does not provide a mechanism for assigning input events to applications without conflicts, nor does it provide a mechanism for transferring data between applications from different developers. Forrest [Reiling. 2014] extended windowing systems to three dimensions to absorb the differences among various virtual reality (VR) devices and realized an environment where multiple applications could be displayed concurrently. However, he did not deal with data transfer between applications, and thus his Wayland compositor was not adequate for a multitasking environment.

Currently, the use of immersive environments is diverse and not limited to games. Especially when used as an assistant for daily life or as a work environment, a new paradigm is needed, like the windows, icons, menus, pointer (WIMP) paradigm in 2D desktop environments, in which users can select and install 3D applications from multiple developers and use them simultaneously in coordination to create a space that suits their style and work. In addition, many people cannot conduct their daily lives and work without being able to use existing 2D applications, so it is necessary to be able to use these applications cooperatively with 3D applications in a 3D space. We designed this new paradigm and developed ZIGEN,

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a windowing system for immersive 3D environments to make them realities.

2 SYSTEM DESIGN

2.1 OVERVIEW

ZIGEN redefined the protocol for a three-dimensional windowing system based on the existing Wayland windowing system protocol. We also implemented ZEN, a 3D display server, as a reference implementation of the protocol. An overview of the system is shown in Figure 3. ZEN is an intermediary between hardware (such as mouse, keyboard, and head-mounted display) and multiple 3D applications, and performs three main functions. First, it defines input focus and transfers events from the input devices to an application in focus. Second, it composites rendering contents of multiple applications into a single space and displays them on a head-mounted display. Third, it provides a mechanism for data sharing between applications, such as drag-and-drop. It is important to note that a 3D application in ZIGEN is not like a "conference room" that dominates the user's entire field of view, but rather an object that dominates a portion of the field of view, such as a "whiteboard" or "shared screen".

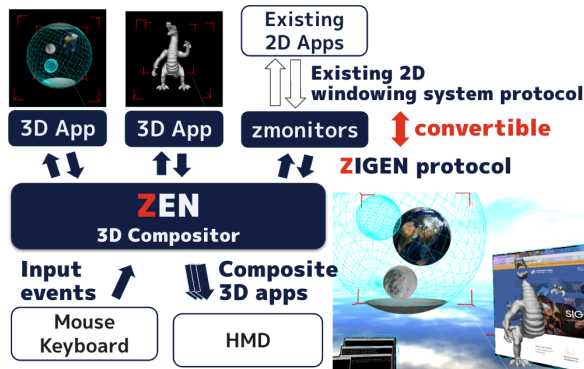


Figure 3: ZEN appropriately distributes events from input devices to 3D applications and composites the rendering content of the 3D applications into a single space that is then displayed on the head-mounted display. *Zmonitors* converts between an existing 2D windowing system protocol and the ZIGEN protocol, so that existing 2D application functions, such as drag-and-drop, can be used in the 3D space. To achieve this, the ZIGEN protocol is designed to be convertible with existing windowing system protocols.

2.2 COMPOSITING 3D APPLICATIONS

ZIGEN adopts Forrest's concept of a cuboid window [Reiling, 2014], a box-shaped region allocated to each 3D application in which the application can draw its 3D content. However, the implementation of the concept of cuboid window differs from Forrest's proposal.

In Forrest's proposal, the display server informs each application of its cuboid window boundaries when they change, and also informs the view-projection matrix obtained from the head-mounted display every frame. The application uses this information to execute the rendering pipeline to create color and depth buffers for

both eyes in which its own content is rendered. The application tells these buffers back to the display server each frame, and the content is displayed on the head-mounted display.

In ZIGEN, the rendering-related part of its protocol is designed following the OpenGL API. Applications transfer vertex buffers, element array buffers, vertex attributes, textures, shaders, and other parameters to ZEN, and ZEN uses this information to execute the rendering pipeline and draw the applications (Figure 2). An important difference from Forrest's proposal is that the view-projection matrix obtained from the head-mounted display and the model matrices indicating the position and transformation of the cuboid windows are not sent to applications, but are designed so that applications use them in shaders. This means that the data sent by applications are no longer dependent on model-view-projection matrix. Therefore, applications that are stationary in the cuboid window do not need to repeat communication for rendering and do not need to waste computational resources. Also, even if an application hangs up, the model-view-projection matrix can be updated for rendering.

2.3 INTEGRATING 2D APPLICATIONS

To use existing 2D applications unmodified in 3D space, ZIGEN does not mirror displays or windows, but instead places a 2D display server named "zmonitors" between the 2D applications and ZEN, which performs the protocol and data conversion (Figure 3). Therefore, ZIGEN's protocol is designed to be dynamically convertible with existing 2D windowing systems. This allows the full functionality of 2D windowing systems to be utilized, enabling drag-and-drop between 2D and 3D applications (Figure 1), optimization of computational resources, and frame synchronization with head-mounted displays.

3 CONCLUSION

We have designed and developed a new windowing system that utilizes existing 2D windowing system functions while integrating them with 3D computer graphics technology to create a more sophisticated multitasking environment in a 3D virtual space. Having provided an environment for multiple 3D applications to work together, each 3D application on this windowing system will be required to provide only one function with high quality. This is in line with the UNIX philosophy of "Do one thing, and do it well," and will lead to a better world of 3D virtual space.

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